PRODUCT USER MANUAL

For Global Ocean Surface Carbon Product
MULTIOBS_GLO_BGC_CARBON_SURFACE_MYNRT_015_008

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	September 2024	all	Creation of the document for the new product MULTIOBS_GLO_BIO_CARBON_SURFACE_MYNRT_015_008 replacing MULTIOBS_GLO_BIO_CARBON_SURFACE_REP_015_008 in order to account for the addition of the retrospective forecast product	Frédéric Chevallier, Thi-Tuyet-Trang Chau, Marion Gehlen	H.Etienne
2.0	November 2025	all	Name changed from MULTIOBS_GLO_BIO_CARBON_SURFACE_MYNRT_015_008 to MULTIOBS_GLO_BGC_CARBON_SURFACE_MYNRT_015_008. Alignment of the content of NRT files on that of MY files New template	Frédéric Chevallier	H.Etienne

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GLOSSARY AND ABBREVIATIONS

CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
NRT	Near Real Time

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the Copernicus Marine Global Ocean Surface Carbon Product **MULTIOBS_GLO_BGC_CARBON_SURFACE_MYNRT_015_008**. The product provides a reconstruction of key variables of the global surface ocean carbonate system from 1985 at a monthly resolution and 0.25°×0.25° spatial resolution.
This product is global. It is defined on a standard grid at 0.25° degree.
The product is organised into two datasets.
The first and main dataset is **cmems_obs-mob_glo_bgc-car_my_irr-i.** It contains seven variables and their associated uncertainty (under the form of a standard deviations):

- **spco2**: surface ocean carbon dioxide partial pressure,
- **spco2_uncertainty**: uncertainty of surface ocean carbon dioxide partial pressure,
- **fgco2**: surface ocean downward mass flux of carbon dioxide expressed as carbon,
- **fgco2_uncertainty**: uncertainty of surface ocean downward mass flux of carbon dioxide expressed as carbon,
- **ph**: surface ocean pH on total scale,
- **ph_uncertainty**: uncertainty of surface ocean pH on total scale,
- **talk:** total alkalinity in surface seawater,
- **talk_uncertainty:** uncertainty of total alkalinity in surface seawater,
- **tco2:** surface ocean dissolved inorganic carbon,
- **tco2_uncertainty:** uncertainty of surface ocean dissolved inorganic carbon,
- **omega_ca:** saturation state for surface seawater with respect to calcite,
- **omega_ca_uncertainty:** uncertainty of saturation state for surface seawater with respect to calcite,
- **omega_ar:** saturation state for surface seawater with respect to aragonite,
- **omega_ar_uncertainty:** uncertainty of saturation state for surface seawater with respect to aragonite.

It is updated with a yearly frequency.
For the recent past, in-between two updates of the MY dataset, **cmems_obs-mob_glo_bgc-car_nrt_irr-i** fills the gap for the same variables.
The product is published on the Copernicus Marine dissemination server after automatic and human quality controls. Files downloaded are in NetCDF format and follow CF-1.6 convention.
The processing system is described in the Quality Information Document.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Table 1 : MOB-TAC MULTIOBS_GLO_BIO_CARBON_SURFACE_MYNRT_015_008

Product Lines	MULTIOBS_GLO_CARBON_SURFACE_MYNRT_015_008
Geographical coverage	Global
Variables	Surface ocean carbon dioxide partial pressure, **spco2** (micro atm) Uncertainty (standard deviation) of surface ocean carbon dioxide partial pressure, **spco2_uncertainty** (micro atm) Surface ocean downward mass flux of carbon dioxide expressed as carbon, **fgco2** (mol C m-2 yr-1) Uncertainty (standard deviation) of surface ocean downward mass flux of carbon dioxide expressed as carbon, **fgco2_uncertainty** (mol C m-2 yr-1) Surface ocean pH on total scale, **ph** (pH units) Uncertainty (standard deviation) of surface ocean pH on total scale, **ph_uncertainty** (pH units) Total alkalinity in surface seawater, **talk** (micro mol kg-1) Uncertainty (standard deviation) of total alkalinity in surface seawater, **talk_uncertainty** (micro mol kg-1) Surface ocean dissolved inorganic carbon, **tco2** (micro mol kg-1) Uncertainty (standard deviation) of surface ocean dissolved inorganic carbon, **tco2_uncertainty** (micro mol kg-1) Saturation state for surface seawater with respect to calcite, **omega_ca** (-) Uncertainty (standard deviation) of saturation state for surface seawater with respect to calcite, **omega_ca_uncertainty** (-) Saturation state for surface seawater with respect to aragonite, **omega_ar** (-) Uncertainty (standard deviation) of saturation state for surface seawater with respect to aragonite, **omega_ar_uncertainty** (-)
Product Type	Near Real Time and Analysis
Analysis	Retrospective analysis for the period from 1985 until December 1 to 2 years ago
Forecast	Retrospective forecast for the period after the analysis.
Available time series	From January 1985
Update frequency	Yearly for the retrospective analysis, monthly for the retrospective forecast
Temporal resolution	Monthly
Update Frequency	Monthly

Target delivery time	
Horizontal resolution	0.25°×0.25°
Number of vertical levels	1
Format	NetCDF CF1.7

b) Production System Description

Detailed information on the systems and products MULTIOBS_GLO_BGC_CARBON_SURFACE_MYNRT_015_008 is on the Copernicus Marine website and in the Quality Information Document.

c) Dataset details

Table 2 : list of the datasets and variable names and unit for the MULTIOBS_GLO _CARBON_SURFACE_MYNRT_015_008

Product Lines	***MULTIOBS_GLO_BGC_CARBON_SURFACE_REP_015_008***	
Geographical coverage	*Global*	
Datasets: ***cmems_obs-mob_glo_bgc-car_my_irr-i*** ***cmems_obs-mob_glo_bgc-car_nrt_irr-i***		
Product Type	*Analysis*	*NRT*
Update frequency	*Yearly*	*Monthly*
Available time series	*From 1985 to -2Y*	*-2Y to -1M*
Target delivery time	*N/A*	*The 20th of the following month at 12:00 UTC*
Temporal resolution	*Monthly mean*	

Dataset: **cmems_obs-mob_glo_bgc-car_my_irr-i** **cmems_obs-mob_glo_bgc-car_nrt_irr-i**
spco2 [micro atm] Surface aqueous partial pressure of CO2 surface_partial_pressure_of_carbon_dioxide_in_sea_water **spco2_uncertainty** [micro atm] Uncertainty (standard deviation) of surface aqueous partial pressure of CO2 uncertainty_surface_partial_pressure_of_carbon_dioxide_in_sea_water
fgco2 [mol C m-2 yr-1]

Surface downward flux of total CO2 surface_downward_mass_flux_of_carbon_dioxide_expressed_as_carbon **fgco2_uncertainty** [mol C m-2 yr-1] Uncertainty (standard deviation) of surface downward flux of total CO2 uncertainty_surface_downward_mass_flux_of_carbon_dioxide_expressed_as_carbon
ph [1] pH on total scale sea_water_ph_reported_on_total_scale **ph_uncertainty** [1] Uncertainty (standard deviation) of pH on total scale uncertainty_sea_water_ph_reported_on_total_scale
talk [micro mol kg-1] Total alkalinity in surface seawater total_alkalinity_in sea_water **talk_uncertainty** [micro mol kg-1] Uncertainty (standard deviation) of total alkalinity in surface seawater uncertainty_total_alkalinity_in sea_water
tco2 [micro mol kg-1] Surface ocean dissolved inorganic carbon dissolved_inorganic_carbon_in_sea_water **tco2_uncertainty** [micro mol kg-1] Uncertainty (standard deviation) of surface ocean dissolved inorganic carbon uncertainty_dissolved_inorganic_carbon_in_sea_water
omega_ca [-] Saturation state for surface seawater with respect to calcite calcite_saturation_state_in_sea_water **omega_ca_uncertainty** [-] Uncertainty (standard deviation) of saturation state for surface seawater with respect to calcite uncertainty_calcite_saturation_state_in_sea_water
omega_ar [-] Saturation state for surface seawater with respect to aragonite aragonite_saturation_state_in_sea_water **omega_ar_uncertainty** [-] Uncertainty (standard deviation) of saturation state for surface seawater with respect to aragonite uncertainty_aragonite_saturation_state_in_sea_water

d) Processing information

Product disseminating schedule :

Type of product	Update	Temporal coverage	Dissemination day
MY	Yearly	1985 to (-24M to -12M)	/
NRT	Monthly	-24M to -1M	(19th of the following month)

e) Catalogue updating

Every month, the NRT delivers the previous month of the surface carbon product.
Every year, the MY dataset is reprocessed and extended with one year. The one year overlapping MY and NRT time serry is delated in the NRT dataset.

f) Time averaging

Dataset frequency	Time averaging
montlhy	monthly means over the calendar month (first to last day of the month)

g) Product use and impact

MULTIOBS_GLO_BGC_CARBON_SURFACE_MYNRT_015_008 provides a consistent suite of variables about the surface ocean carbonate system. It is driven by sparse CO2 fugacity measurements made since 1985 that have been spatialized with related observation datasets, mainly derived from satellite measurements. MULTIOBS_GLO_BGC_CARBON_SURFACE_MYNRT_015_008 is therefore primarily an observation-based data product with no physical model contribution at the level of the fugacity maps and limited one when translating them into air-sea fluxes and then to pH and other variables of the surface ocean carbonate system.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

5) FILE SIZE

DATASET NAME	FILE NAME	DIMENSION [MB]	
		Compressed	Uncompressed
cmems_obs-mob_glo_bgc-car_my_irr-i	cmems_obs-mob_glo_bgc-car_$dd_irr_i_$date1$T0000Z_P$date2$T1545Z.nc	27	55
cmems_obs-mob_glo_bgc-car_nrt_irr-i	cmems_obs-mob_glo_bgc-car_$dd_irr_i_$date1$T0000Z_P$date2$T1545Z.nc	4	8

6) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

7) REFERENCES

See the Quality Information Document (QUID):
https://documentation.marine.copernicus.eu/QUID/CMEMS-MOB-QUID-015-008.pdf

8) ANNEX

a) Structure and semantic of NetCDF maps files

An example output NetCDF file header for cmems_obs-mob_glo_bgc-car_my_irr_i is inserted below:

```
dimensions:
	time = UNLIMITED ; // (1 currently)
	longitude = 1440 ;
	latitude = 713 ;
variables:
	float time(time) ;
		time:standard_name = "time" ;
		time:long_name = "time" ;
		time:units = "hours since 1950-01-01 00:00:00" ;
		time:calendar = "gregorian" ;
		time:axis = "T" ;
	float longitude(longitude) ;
		longitude:standard_name = "longitude" ;
		longitude:long_name = "longitude" ;
		longitude:units = "degrees_east" ;
		longitude:axis = "X" ;
	float latitude(latitude) ;
		latitude:standard_name = "latitude" ;
		latitude:long_name = "latitude" ;
		latitude:units = "degrees_north" ;
		latitude:axis = "Y" ;
	float spco2(time, latitude, longitude) ;
		spco2:standard_name = "surface_partial_pressure_of_carbon_dioxide_in_sea_water" ;
		spco2:long_name = "Surface aqueous partial pressure of CO2" ;
		spco2:units = "micro atm" ;
		spco2:_FillValue = 9.96921e+36f ;
		spco2:missing_value = 9.96921e+36f ;
	float spco2_uncertainty(time, latitude, longitude) ;
		spco2_uncertainty:standard_name =
"uncertainty_surface_partial_pressure_of_carbon_dioxide_in_sea_water" ;
		spco2_uncertainty:long_name = "Uncertainties (standard deviation) of surface
aqueous partial pressure of CO2" ;
		spco2_uncertainty:units = "micro atm" ;
		spco2_uncertainty:_FillValue = 9.96921e+36f ;
		spco2_uncertainty:missing_value = 9.96921e+36f ;
	float fgco2(time, latitude, longitude) ;
		fgco2:standard_name =
"surface_downward_mass_flux_of_carbon_dioxide_expressed_as_carbon" ;
		fgco2:long_name = "Surface downward flux of total CO2" ;
		fgco2:units = "molC m-2 yr-1" ;
		fgco2:_FillValue = 9.96921e+36f ;
		fgco2:missing_value = 9.96921e+36f ;
	float fgco2_uncertainty(time, latitude, longitude) ;
		fgco2_uncertainty:standard_name =
"uncertainty_surface_downward_mass_flux_of_carbon_dioxide_expressed_as_carbon" ;
		fgco2_uncertainty:long_name = "Uncertainties (standard deviation) of surface
downward flux of total CO2" ;
```

```
            fgco2_uncertainty:units = "molC m-2 yr-1" ;
            fgco2_uncertainty:_FillValue = 9.96921e+36f ;
            fgco2_uncertainty:missing_value = 9.96921e+36f ;
      float ph(time, latitude, longitude) ;
            ph:standard_name = "sea_water_ph_reported_on_total_scale" ;
            ph:long_name = "pH on total scale" ;
            ph:units = "1" ;
            ph:_FillValue = 9.96921e+36f ;
            ph:missing_value = 9.96921e+36f ;
      float ph_uncertainty(time, latitude, longitude) ;
            ph_uncertainty:standard_name =
"uncertainty_sea_water_ph_reported_on_total_scale" ;
            ph_uncertainty:long_name = "Uncertainties (standard deviation) of pH on total scale"
;
            ph_uncertainty:units = "1" ;
            ph_uncertainty:_FillValue = 9.96921e+36f ;
            ph_uncertainty:missing_value = 9.96921e+36f ;
      float talk(time, latitude, longitude) ;
            talk:standard_name = "total_alkalinity_in_sea_water" ;
            talk:long_name = "Total alkalinity in surface seawater" ;
            talk:units = "micro mol kg-1" ;
            talk:_FillValue = 9.96921e+36f ;
            talk:missing_value = 9.96921e+36f ;
      float talk_uncertainty(time, latitude, longitude) ;
            talk_uncertainty:standard_name = "uncertainty_total_alkalinity_in_sea_water" ;
            talk_uncertainty:long_name = "Uncertainty (standard deviation) of total alkalinity in
surface seawater" ;
            talk_uncertainty:units = "micro mol kg-1" ;
            talk_uncertainty:_FillValue = 9.96921e+36f ;
            talk_uncertainty:missing_value = 9.96921e+36f ;
      float tco2(time, latitude, longitude) ;
            tco2:standard_name = "dissolved_inorganic_carbon_in_sea_water" ;
            tco2:long_name = "Surface ocean dissolved inorganic carbon" ;
            tco2:units = "micro mol kg-1" ;
            tco2:_FillValue = 9.96921e+36f ;
            tco2:missing_value = 9.96921e+36f ;
      float tco2_uncertainty(time, latitude, longitude) ;
            tco2_uncertainty:standard_name =
"uncertainty_dissolved_inorganic_carbon_in_sea_water" ;
            tco2_uncertainty:long_name = "Uncertainty (standard deviation) of surface ocean
dissolved inorganic carbon" ;
            tco2_uncertainty:units = "micro mol kg-1" ;
            tco2_uncertainty:_FillValue = 9.96921e+36f ;
            tco2_uncertainty:missing_value = 9.96921e+36f ;
      float omega_ca(time, latitude, longitude) ;
            omega_ca:standard_name = "calcite_saturation_state_in_sea_water" ;
            omega_ca:long_name = "Saturation state for surface seawater with respect to
calcite" ;
            omega_ca:units = "-" ;
            omega_ca:_FillValue = 9.96921e+36f ;
            omega_ca:missing_value = 9.96921e+36f ;
      float omega_ca_uncertainty(time, latitude, longitude) ;
            omega_ca_uncertainty:standard_name =
"uncertainty_calcite_saturation_state_in_sea_water" ;
```

```
            omega_ca_uncertainty:long_name = "Uncertainty (standard deviation) of saturation
state for surface seawater with respect to calcite" ;
            omega_ca_uncertainty:units = "-" ;
            omega_ca_uncertainty:_FillValue = 9.96921e+36f ;
            omega_ca_uncertainty:missing_value = 9.96921e+36f ;
      float omega_ar(time, latitude, longitude) ;
            omega_ar:standard_name = "aragonite_saturation_state_in_sea_water" ;
            omega_ar:long_name = "Saturation state for surface seawater with respect to
aragonite" ;
            omega_ar:units = "-" ;
            omega_ar:_FillValue = 9.96921e+36f ;
            omega_ar:missing_value = 9.96921e+36f ;
      float omega_ar_uncertainty(time, latitude, longitude) ;
            omega_ar_uncertainty:standard_name =
"uncertainty_aragonite_saturation_state_in_sea_wate" ;
            omega_ar_uncertainty:long_name = "Uncertainty (standard deviation) of saturation
state for surface seawater with respect to aragonite" ;
            omega_ar_uncertainty:units = "-" ;
            omega_ar_uncertainty:_FillValue = 9.96921e+36f ;
            omega_ar_uncertainty:missing_value = 9.96921e+36f ;

// global attributes:
            :CDI = "Climate Data Interface version 2.0.6 (https://mpimet.mpg.de/cdi)" ;
            :Conventions = "CF-1.6" ;
            :_NCProperties = "version=2,netcdf=4.7.4,hdf5=1.10.7," ;
            :institution = "LSCE/CEA" ;
            :title = "Surface ocean carbon fields" ;
            :description = "MULTIOBS Surface carbon fields" ;
            :domain_name = "GLO" ;
            :references = "Chau, T. T. T., Gehlen, M., and Chevallier, F.: A seamless ensemble-
based reconstruction of surface ocean pCO2 and air–sea CO2 fluxes over the global coastal
and open oceans, Biogeosciences, 19, 1087–1109, https://doi.org/10.5194/bg-19-1087-
2022, 2022." ;
            :history = "Thu Aug 31 17:43:34 2023: cdo -selyear,2013 -selmon,09 dataset-carbon-rep-
monthly_1985to2022.nc ../Monthly_data/dataset-carbon-rep-
monthly_20130915T0000Z_P20230821T1545Z.nc" ;
            :CDO = "Climate Data Operators version 2.0.6 (https://mpimet.mpg.de/cdo)" ;
}
```